SwiGLU as a Routed CP Tensor Model

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Abstract

Modern transformers spend much of their parameter budget in SwiGLU feedforward blocks, but the structural bias of these blocks is less understood than that of attention. We show that SwiGLU admits an exact input-routed CP decomposition of a third-order interaction tensor field: each hidden unit contributes a fixed rank-one atom, routed by an input-dependent sigmoid gate. This view exposes a diagonal latent-core restriction, in which only same-index main and gate features interact. We relax this restriction with a Tucker-core FFN and prove an exact aligned-width separation: for fixed latent dictionaries, representing a generic Tucker-core block with latent dimensions (r, s) requires $r\ \min(r, s)$ SwiGLU units. Empirically, constant-gate ablations severely degrade pretrained Qwen models, synthetic teacher-student experiments recover the predicted width threshold, and post-hoc diagonalization of a trained Tucker-core LM increases validation perplexity by $518\times$. Together, these results identify SwiGLU as a routed low-rank tensor model with a concrete diagonal bottleneck.

1. Introduction

Attention admits multiple complementary structural interpretations: as Nadaraya–Watson kernel regression (Nadaraya, 1964; Pai, 2025), as a Boltzmann distribution over key-value pairs, and as a routing mechanism with temperature $\beta = \tau^{-1}$ that interpolates between diffuse averaging and sharp retrieval. The SwiGLU feedforward network (FFN) block (Shazeer, 2020), which holds the majority of a modern transformer's parameters and empirically outperforms ReLU/GELU FFNs at matched scale, lacks an analogous account of its structural bias. We apply tensor methods to fill this gap.

At a fixed token position, let $x \in \mathbb{R}^d$ denote the residual stream entering the FFN. SwiGLU computes

$$y = U^\top h, \qquad h = (W^\top x) \odot \mathrm{SiLU}(G^\top x), \tag{1}$$

with $W, G \in \mathbb{R}^{d\times m}$ and $U \in \mathbb{R}^{m\times d}$, so that $U^\top \in \mathbb{R}^{d\times m}$ shares its shape with W and G; in the Tucker recovery setting of Section 3, the output factor takes the form $R = U^\top$. Section 2 shows that this map is an exact input-routed CP decomposition of a third-order interaction tensor and that its same-index channel coupling is a structural restriction rather than an arbitrary parameterization choice; Section 3 lifts this restriction via a Tucker-factorized generalization that recovers SwiGLU as a degenerate special case. Section 4 states the resulting separation theorem; Section 5 tests whether pretrained SwiGLU models use the routed-CP structure and whether trained Tucker LMs use non-diagonal core structure that the aligned theorem identifies as costly.

2. SwiGLU as a Routed CP Decomposition

Writing the j-th columns of W and G as $w_j, g_j \in \mathbb{R}^d$, the j-th hidden coordinate of (1) is $h_j(x) = (w_j^\top x)\,\mathrm{SiLU}(g_j^\top x)$. The SiLU nonlinearity has an exact multiplicative factorization,

$$\mathrm{SiLU}(z) = z\,\sigma(z), \qquad \sigma(z) = (1 + e^{-z})^{-1}, \tag{2}$$

so each hidden coordinate decomposes as

$$h_j(x) = \underbrace{(w_j^\top x)(g_j^\top x)}_{\text{rank-one bilinear interaction}} \cdot \underbrace{\sigma(g_j^\top x)}_{\alpha_j(x)}. \tag{3}$$

Each unit thus realizes a rank-one quadratic form $x^\top(w_j g_j^\top)x$ between two learned directions, modulated by an input-dependent scalar gate $\alpha_j(x) \in [0, 1]$.

Tensor form. Define the input-dependent interaction tensor $\mathcal{A}(x) \in \mathbb{R}^{d\times d\times d}$ by $y_o = \sum_{a,b} \mathcal{A}_{oab}(x)\, x_a x_b$. Substituting (3) into the output projection yields

$$\mathcal{A}(x) = \sum_{j=1}^{m} \alpha_j(x)\, u_j \otimes w_j \otimes g_j, \tag{4}$$

where u_j is the j-th column of $U^\top$ and $\alpha_j(x) = \sigma(g_j^\top x)$ is the *routing coefficient* that gates the j-th rank-one atom. We

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emphasise that $\mathcal{A}(x)$ is an input-dependent *tensor field*, not a fixed third-order tensor: (4) is a canonical polyadic (CP) decomposition (Kolda & Bader, 2009) of each $\mathcal{A}(x)$ with shared factor matrices $(U^\top, W, G)$ and input-dependent mixing weights $\alpha_j(x)$, exact and pointwise in x. The two bilinear modes of $\mathcal{A}(x)$ are functionally symmetric under $(a, b) \to (b, a)$; the gate direction g_j breaks this symmetry by entering α_j.

Same-index restriction. The CP form (4) couples w_j only with g_j and u_j, never with w_k, g_k, or u_k for $k \neq j$. SwiGLU therefore realizes multiplicative feature mixing (Jayakumar et al., 2020) only through sums of rank-one separable factors with diagonal index pairing. Jayakumar et al. (2020) frame gating, attention, and hypernetworks as instances of multiplicative interaction; our contribution is the specific routed-CP / Tucker factorization the SwiGLU block admits and the aligned-width separation it implies. Equivalently: writing the layer as a Tucker decomposition (Kolda & Bader, 2009) with hidden dimension m and $r = s = m$, the core tensor must be superdiagonal, $\mathcal{C}_{oij} = \delta_{oi}\delta_{ij}$. Pearce et al. (2025) analyze *trained* bilinear MLPs (the $\alpha_j \equiv 1$ case) via eigendecomposition; the routed-CP form makes the gating part of the structural picture and reveals the diagonal restriction as a condition on the latent core, not on the residual stream.

3. Tucker-Core FFN

The natural relaxation of (4) is to drop the superdiagonal constraint on the latent core. We define the *Tucker-core FFN* by latent projections $P, Q \in \mathbb{R}^{d\times r}$, output projection $R \in \mathbb{R}^{d\times s}$, and a learned core $\mathcal{C} \in \mathbb{R}^{s\times r\times r}$. For $p = P^\top x$ and $q = Q^\top x$, the block computes

$$z_o = \sum_{i,j=1}^{r} \mathcal{C}_{oij}\, p_i\, \mathrm{SiLU}(q_j), \qquad y = Rz. \tag{5}$$

The same SiLU identity gives $z_o = \sum_{ij} \mathcal{C}_{oij}\, \sigma(q_j)\, p_i\, q_j$, so the block still realizes an input-routed third-order interaction tensor; the routing is now controlled by the latent gate vector q rather than by per-channel $\sigma(g_j^\top x)$, and cross-index coupling $\mathcal{C}_{oij}$ with $i \neq j$ is permitted. SwiGLU is recovered as the special case $r = s = m$ with $\mathcal{C}_{oij} = \delta_{oi}\delta_{ij}$, in which (5) reduces to $z_i = p_i\, \mathrm{SiLU}(q_i)$ (Lemma A.1). The block has parameter count $2dr + ds + sr^2$, against $3dm$ for SwiGLU; relative to SwiGLU, it is a strictly larger function class as a parameterized object, but a parameter-budget comparison is conditional on (d, r, s) and is given in Appendix A (§A.4).

Positioning. Several lines of prior tensor work on neural networks address related but distinct questions. Cohen et al. (2016) establish CP-vs-hierarchical-Tucker expressivity gaps for deep convolutional arithmetic circuits; Novikov et al. (2015) compress weight matrices via tensor-train factorizations; classical factorization machines (Rendle, 2010) model pairwise feature interactions through low-rank multiplicative atoms. Our object is the single transformer-FFN block, read as a routed third-order interaction with neither weight compression nor depth as the lever; the contribution is the local diagonal-vs-dense Tucker analogue and an exact width separation in the aligned class.

4. The Diagonal Bottleneck

We isolate the structural cost of the same-index restriction. Holding the latent feature spaces P, Q fixed, we ask how many SwiGLU channels are required to realize the same map as a Tucker-core block. This is not a universal lower bound against arbitrary SwiGLU dictionaries; it isolates the diagonal pairing constraint from the cost of learning the latent dictionaries themselves.

Definition 4.1 (Aligned SwiGLU representation)**.** Given Tucker latent matrices $P, Q \in \mathbb{R}^{d\times r}$, an *aligned width-$m$ SwiGLU representation* is a map

$$\widehat{y}(x) = \sum_{j=1}^{r}\sum_{\nu=1}^{k_j} u_{j\nu}\, (a_{j\nu}^\top p)\, \mathrm{SiLU}(q_j), \qquad \sum_{j=1}^{r} k_j = m, \tag{6}$$

with $u_{j\nu} \in \mathbb{R}^d$, $a_{j\nu} \in \mathbb{R}^r$, $p = P^\top x$, and $q = Q^\top x$. Equivalently, each unit uses a main vector in $\mathrm{span}(P)$ and one of the gate vectors $Q_{:j}$.

The aligned class is the natural object of study whenever the latent dictionaries (P, Q) are already learned: distillation from a Tucker teacher, post-training compression, and the inner-loop alignment a SwiGLU would have to discover before exploiting the Tucker structure. In each case, the question reduces to how many SwiGLU units are needed *given those dictionaries*. Theorem 4.2 is the answer to that question; it does not lower-bound an unconstrained SwiGLU that is free to learn its own (P, Q).

Grouping the contributions in (5) by $\mathrm{SiLU}(q_j)$ gives

$$y_{\text{Tucker}}(x) = \sum_{j=1}^{r} V_j\, p\, \mathrm{SiLU}(q_j), \qquad V_j := R\,\mathcal{C}_{:,:,j} \in \mathbb{R}^{d\times r}. \tag{7}$$

We call V_j the *output-by-main-feature coefficient matrix* attached to gate q_j. The aligned SwiGLU map admits the same decomposition with V_j replaced by $\widehat{V}_j := \sum_{\nu=1}^{k_j} u_{j\nu} a_{j\nu}^\top$, which satisfies $\mathrm{rank}(\widehat{V}_j) \leq k_j$.

Theorem 4.2 (Diagonal bottleneck)**.** *Assume $[P\ \ Q] \in \mathbb{R}^{d\times 2r}$ has full column rank. Then the minimum width of an aligned SwiGLU representation that exactly realizes a Tucker-core FFN with output-coefficient matrices $\{V_j\}_{j=1}^{r}$*

is

$$m_{\min} = \sum_{j=1}^{r} \operatorname{rank}(V_j). \tag{8}$$

The lower bound follows by varying (p, q) independently, possible by the rank assumption on $[P \ \ Q]$, and using $\mathrm{SiLU}(0) = 0$ together with $\mathrm{SiLU}(1) \neq 0$ to isolate each gate. The upper bound is constructive: a rank decomposition $V_j = \sum_{\nu=1}^{\rho_j} u_{j\nu} a_{j\nu}^\top$ realizes the j-th gated slice using $\rho_j = \operatorname{rank}(V_j)$ aligned SwiGLU units. Full proof in Appendix A.

Corollary 4.3 (Generic separation)**.** *If R has full column rank and each slice $\mathcal{C}_{:,:,j} \in \mathbb{R}^{s\times r}$ is generic, then* $\operatorname{rank}(V_j) = \min(r, s)$ *for every j, so $m_{\min} = r\min(r, s)$. In particular, for the balanced case $r = s = k$,*

$$m_{\min} = k^2, \tag{9}$$

so a generic Tucker-core FFN with $3dk + k^3$ parameters requires an aligned SwiGLU block with $3dk^2$ parameters to simulate exactly; for $d \gg k^2$ this is a factor-k separation.

Remark 4.4. The bound is representational. It says nothing about whether SGD recovers the optimal Tucker core or whether stacked SwiGLU layers can compensate at matched depth, both of which we return to in Section 6. Numerical-rank intuition for approximate simulation is sketched in Appendix A (§A.5); a complete approximation theorem is left to future work.

5. Empirical Characterization

The CP decomposition of (4) is algebraically exact, but exactness alone does not establish that the structure is informative: if every $\alpha_j(x)$ were approximately constant, the decomposition would be correct but inert. We test whether trained SwiGLU models actually use the routed-CP structure, whether the aligned width separation of Theorem 4.2 is reachable in practice, and whether trained Tucker LMs use non-diagonal core structure that an aligned SwiGLU at width r would be unable to remove.

5.1. Post-hoc constant-α ablation breaks Qwen2.5-0.5B SwiGLU

We ablate the input-dependence of the gate by replacing $\alpha_j(x) = \sigma(g_j^\top x)$ with a constant α_0 at inference time, turning each hidden unit into $h_j(x) = \alpha_0\,(w_j^\top x)(g_j^\top x)$. We use Qwen2.5-0.5B (Yang et al., 2024) (24 SwiGLU layers, $d = 896$, $m = 4864$) on a 4096-token chunk of WikiText-2. Three replacements are tested: $\alpha_0 = 0.5$ (uniform, near the early-layer mean), $\alpha_0 = \mathbb{E}_x[\sigma(g_j^\top x)]$ (per-channel mean, preserving average scale per channel), and $\alpha_0 = 1$ (the static bilinear setting of Pearce et al. (2025)). All three apply to every layer simultaneously; perplexity is evaluated on a held-out chunk not used to calibrate the per-channel means.

Replacement	Perplexity
Baseline (full routing)	16.8
Uniform ($\alpha = 0.5$)	2.7×10^5
Bilinear ($\alpha = 1$)	7.4×10^5
Per-channel mean ($\alpha = \mathbb{E}[\alpha_j]$)	6.4×10^6

Table 1. Post-hoc constant-α replacement on a pretrained Qwen2.5-0.5B SwiGLU. All three replacements degrade perplexity by 4–6 orders of magnitude.

Every constant replacement we tested catastrophically degrades the model's outputs. The ordering is informative: the per-channel mean is the most damaging despite preserving each channel's average scale, suggesting that preserving per-channel average scale is insufficient and that rare or input-specific gate deviations may be important. The $\alpha \equiv 1$ ablation corresponds algebraically to the bilinear GLU form analysed by Pearce et al. (2025) on trained bilinear MLPs; imposing it post hoc on a trained SwiGLU is a different setting and is catastrophic here. The table shows that pretrained Qwen2.5-0.5B SwiGLU layers rely strongly on their learned input-dependent routing, and that post-hoc constant-α replacements are not function-preserving.

5.2. Synthetic verification of Theorem 4.2

To check that the lower bound (8) is tight in practice and not just in algebra, we generate a synthetic Tucker-core teacher with full-rank generic frontal slices ($d = 64$, balanced $r = s = k$ for $k \in \{4, 8, 16\}$, $\mathcal{C}_{:,:,j}$ resampled until full rank) and fit three student families to its outputs by mean-squared error: an *unconstrained* SwiGLU with free $w_l, g_l \in \mathbb{R}^d$, an *aligned* SwiGLU restricted to the matched-coordinates hypothesis class of Definition 4.1 ($w_l = Pa_l$, $g_l = Q_{:,j_l}$), and a Tucker-core control. We sweep the SwiGLU width m across a geometric grid bracketing k^2 and report the best validation MSE over 8 random seeds and 8000 Adam steps with a cosine schedule.

The aligned-SwiGLU curve (Figure 1, dashed red) shows the predicted knee at $m = k^2$ for every k: validation MSE drops by roughly twelve orders of magnitude between $m < k^2$ and $m \geq 2k^2$, reaching 1.3×10^{-14} at $k = 4$, 6.4×10^{-14} at $k = 8$, and 3.4×10^{-12} at $k = 16$; the SVD-constructed point at $m = k^2$ in Figure 1 confirms that the upper bound is attained without optimization. The unconstrained SwiGLU is a strict superset of the aligned class, so its persistent plateau ($\sim 10^{-4}$ at $k = 4$, $\sim 10^{-3}$ at $k = 8$, $\sim 10^{-1}$ at $k = 16$) is an optimization failure, not a representational one: the unconstrained student did not discover the aligned construction under this optimizer in the swept range, indicating that the constructive representation guaranteed by Theorem 4.2 is not automatically found by naive end-to-end

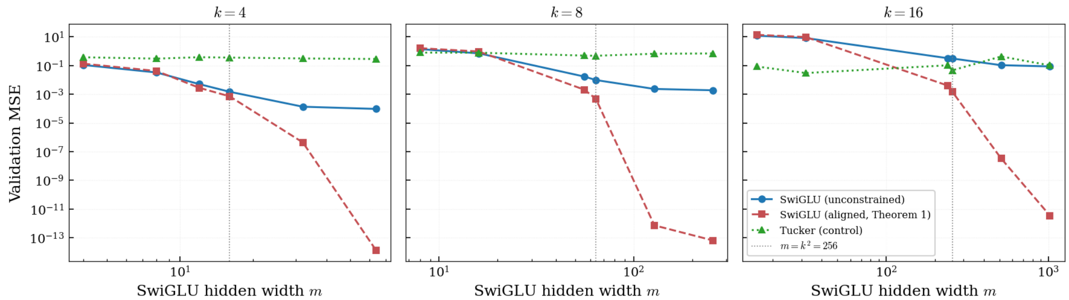

Figure 1. Synthetic verification of Theorem 4.2. Validation MSE of student SwiGLU and Tucker FFNs fit to a generic Tucker teacher with full-rank cores ($d = 64$, balanced $r = s = k$ for $k \in \{4, 8, 16\}$, eight seeds, 8000 Adam steps with cosine schedule). Aligned SwiGLU (matched-coordinates hypothesis class of Definition 4.1) shows the predicted knee at $m = k^2$ and reaches machine precision at $m \geq 2k^2$ in every panel; unconstrained SwiGLU plateaus far above machine precision over the same range under the same optimizer. Vertical dotted line marks $m = k^2$.

training. The aligned hypothesis class makes the bound reachable in practice. The Tucker control's persistent floor reflects optimization, not capacity.

5.3. A trained Tucker LM uses non-diagonal core structure that cannot be removed post hoc

The synthetic experiment verifies the bound at machine precision in the matched-coordinates setting that the theorem assumes. We now ask whether a trained Tucker LM uses core structure that the SwiGLU recovery condition would force it to discard. We train a Tucker-core transformer (d=512, L=8, r=s=128, $\sim$ 50M parameters) on 100M tokens of FineWeb-Edu (Penedo et al., 2024) and apply two structural ablations to the trained core $\mathcal{C} \in \mathbb{R}^{s\times r\times r}$ at evaluation time. The λ-sweep interpolates $\mathcal{C}^{(\lambda)} = (1-\lambda)\mathcal{C} + \lambda\,\mathrm{diag}(\mathcal{C})$, where $\mathrm{diag}(\mathcal{C})$ retains only the superdiagonal entries $\mathcal{C}_{ooo}$. The ρ-sweep replaces the output-by-main-feature matrix $V_j = R\,\mathcal{C}^{(j)}$ with its rank-ρ SVD truncation gate-by-gate, exactly the constraint Theorem 4.2 imposes on an aligned SwiGLU of width $m = \rho \cdot r$.

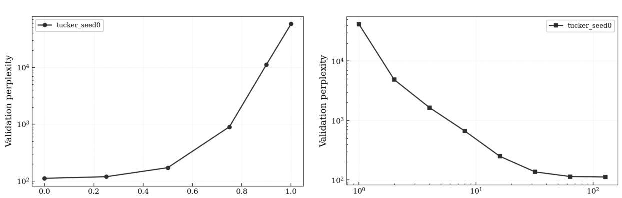

Figure 2. Diagonal-bottleneck cost on a trained Tucker LM ($d = 512$, $r = s = 128$, 100M FineWeb-Edu tokens, 64-sequence held-out validation set). Left: validation perplexity as $C \to \mathrm{diag}(C)$ via $C^{(\lambda)} = (1-\lambda)C + \lambda\,\mathrm{diag}(C)$; the λ=1 point is the SwiGLU recovery condition of Lemma A.1. Right: per-gate SVD truncation $V_j = R\,C^{(j)}$ at rank ρ, equivalent to an aligned SwiGLU of width $m = \rho \cdot r$ (Theorem 4.2). Trained perplexity 111.84 vs. fully diagonal-projected 57 975 (518×); baseline preservation at ρ=128 holds within 0.002% relative error.

Forcing $\mathcal{C}$ onto its superdiagonal, the SwiGLU recovery condition of Lemma A.1, costs a factor of 518× in validation perplexity on a held-out FineWeb-Edu split (111.84 → 57 975; Figure 2, left). The ρ-truncation curve (right) traces the bound directly: ρ=1 recovers an aligned-SwiGLU width-r block and yields perplexity 4.2×10^4; ρ=8 already drops to 667; the curve flattens to baseline (111.84) at ρ=128, with a baseline preservation check passing at 0.002% relative error. Per-gate stable rank of V_j in the trained model concentrates around $\bar{\rho} \approx 4$ (Appendix B.2), an order of magnitude above the rank-one ceiling that aligned SwiGLU at width r would impose. Matched-parameter Tucker students fitted to a Tucker teacher achieve 22% lower MSE than SwiGLU students at the LM-config budget (Appendix B.3), realising the layer-level gap Theorem 4.2 predicts.

6. Discussion

Together, the three experiments characterise the diagonal bottleneck from complementary directions: pretrained SwiGLU layers depend on input-routed gating (§5.1), synthetic teacher–student fits attain the $r \min(r, s)$ width threshold at machine precision (§5.2), and a trained Tucker LM stores cross-channel core structure that an aligned SwiGLU at width r cannot retain (§5.3). The bottleneck is therefore both representationally tight and empirically loaded.

Various open directions remain. Theorem 4.2 is a single-layer aligned result; whether L stacked SwiGLU layers can simulate one Tucker layer at matched compute is the natural follow-up. The bound is representational rather than about learnability (Remark 4.4). End-to-end training at the 50M-parameter scale (Appendix B.5) shows a variance-preserving Tucker initialisation matching SwiGLU within run-to-run noise, but the single-seed result is not strong enough to claim either superiority or equivalence; we leave end-to-end scaling to future work. The bilinear $\alpha_j \equiv 1$ setting analysed by Pearce et al. (2025) via eigendecomposition is recovered as the gate-saturated limit ($\alpha_j(x) \to 1$) of the routed-CP form. Connections to circuit

/ tensor-factorization formalisms (Loconte et al., 2024) and to transcoders (Dunefsky et al., 2024; Paulo et al., 2025) are discussed in Appendix C.

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A. Proof of Theorem 4.2

A.1. Setup

Before turning to the proof, we verify that the Tucker-core FFN strictly contains SwiGLU, so that the bound below measures a genuine cost of the diagonal restriction.

Lemma A.1 (SwiGLU recovery)**.** *Setting $r = s = m$, $P = W$, $Q = G$, $R = U^\top$, and $\mathcal{C}_{oij} = \delta_{oi}\delta_{ij}$ in (5) reduces the Tucker-core FFN to the SwiGLU map (1).*

Proof. With $\mathcal{C}_{oij} = \delta_{oi}\delta_{ij}$, equation (5) collapses to $z_o = p_o\,\mathrm{SiLU}(q_o)$, so $z = p \odot \mathrm{SiLU}(q) = (P^\top x) \odot \mathrm{SiLU}(Q^\top x)$. With $P = W$, $Q = G$, $R = U^\top$, this is exactly h from (1), and $y = Rz = U^\top h$. □

We restate notation for self-containedness. Let $d \geq 2r$, $\phi(t) = \mathrm{SiLU}(t) = t\sigma(t)$ with $\sigma(t) = (1+e^{-t})^{-1}$, and let $\phi(0) = 0$, $\phi(1) \neq 0$ (the only properties of ϕ used below). The Tucker-GLU output, expanded along the gate index, is

$$y_{\mathrm{Tucker}}(x) = \sum_{o=1}^{s}\sum_{i,j=1}^{r} R_{:o}\,\mathcal{C}_{oij}\,p_i\,\phi(q_j) = \sum_{j=1}^{r} V_j\,p\,\phi(q_j), \tag{10}$$

where, defining the j-th core slice $C^{(j)} \in \mathbb{R}^{s\times r}$ by $C^{(j)}_{oi} := \mathcal{C}_{oij}$,

$$V_j := R\,C^{(j)} \in \mathbb{R}^{d\times r}. \tag{11}$$

A width-m aligned SwiGLU representation (6) likewise admits the decomposition $\widehat{y}(x) = \sum_j \widehat{V}_j\,p\,\phi(q_j)$, with

$$\widehat{V}_j := \sum_{\nu=1}^{k_j} u_{j\nu}\,a_{j\nu}^\top \in \mathbb{R}^{d\times r}, \qquad \mathrm{rank}(\widehat{V}_j) \leq k_j. \tag{12}$$

Assumption A.2 (Latent surjectivity)**.** $[P\ Q] \in \mathbb{R}^{d\times 2r}$ has full column rank. Equivalently, $[P\ Q]^\top : \mathbb{R}^d \to \mathbb{R}^{2r}$ has rank $2r$, so its image is all of $\mathbb{R}^{2r} \cong \mathbb{R}^r \times \mathbb{R}^r$, and $x \mapsto (P^\top x, Q^\top x)$ is surjective onto $\mathbb{R}^r \times \mathbb{R}^r$.

A.2. Proof of Theorem 4.2

Lower bound. Suppose an aligned representation (6) satisfies $\widehat{y} \equiv y_{\mathrm{Tucker}}$. By Assumption A.2, (p, q) ranges over all of $\mathbb{R}^r \times \mathbb{R}^r$, so

$$\sum_{j=1}^{r}(V_j - \widehat{V}_j)\,p\,\phi(q_j) = 0 \quad \text{for all } p, q \in \mathbb{R}^r. \tag{13}$$

Fix j_0 and choose $q_{j_0} = 1$, $q_j = 0$ for $j \neq j_0$. Since $\phi(0) = 0$, only the j_0-th term survives in (13):

$$(V_{j_0} - \widehat{V}_{j_0})\,p\,\phi(1) = 0 \quad \text{for all } p \in \mathbb{R}^r.$$

Since $\phi(1) \neq 0$, this forces $V_{j_0} = \widehat{V}_{j_0}$. As j_0 was arbitrary, $V_j = \widehat{V}_j$ for all j. Combining with (12) gives $k_j \geq \mathrm{rank}(V_j)$, and summing over j yields $m \geq \sum_j \mathrm{rank}(V_j)$.

Tightness. Set $\rho_j := \mathrm{rank}(V_j)$ and let $V_j = \sum_{\nu=1}^{\rho_j} u_{j\nu}a_{j\nu}^\top$ be any rank decomposition. Then

$$\begin{aligned} V_j\,p\,\phi(q_j) &= \sum_{\nu=1}^{\rho_j} u_{j\nu}\,(a_{j\nu}^\top p)\,\phi(q_j) \\ &= \sum_{\nu=1}^{\rho_j} u_{j\nu}\,\big((Pa_{j\nu})^\top x\big)\,\phi(Q_{:j}^\top x), \end{aligned}$$

which is a sum of ρ_j aligned SwiGLU units (main vector $Pa_{j\nu}$, gate vector $Q_{:j}$, output vector $u_{j\nu}$). Realizing every j this way gives total width $\sum_j \rho_j$, attaining the bound. □

A.3. Generic-rank corollary

Lemma A.3 (Generic rank)**.** *The set of $s \times r$ matrices with rank strictly less than $\min(r, s)$ is a proper algebraic subvariety of $\mathbb{R}^{s\times r}$, defined by the vanishing of all $\min(r, s) \times \min(r, s)$ minors. In particular, it has Lebesgue measure zero, so a generic $s \times r$ matrix has rank $\min(r, s)$.*

Proof of Theorem 4.3. By (11) and the full column rank of R, $\operatorname{rank}(V_j) = \operatorname{rank}(R\,C^{(j)}) = \operatorname{rank}(C^{(j)})$. By Theorem A.3, $\operatorname{rank}(C^{(j)}) = \min(r, s)$ for each j, so $\operatorname{rank}(V_j) = \min(r, s)$ and Theorem 4.2 gives $m_{\min} = r\ \min(r, s)$. □

Remark A.4 (Almost-sure formulation). If $d \geq s$ and $(R, \mathcal{C})$ are drawn from any distribution absolutely continuous with respect to Lebesgue measure on the parameter space, then by Theorem A.3 the rank-deficient locus has measure zero, so $\operatorname{rank}(V_j) = \min(r, s)$ almost surely.

A.4. Parameter-budget separation

The two parameter counts are $N_{\text{Tucker}}(d, r, s) = 2dr+ds+sr^2$ and $N_{\text{SwiGLU}}(d, m) = 3dm$. Combining with Theorem 4.3 gives the matched-simulation cost $N_{\text{SwiGLU,eq}}(d, r, s) = 3d\,r \min(r, s)$. Tucker is parameter-cheaper than the minimum-width aligned simulator when

$$2dr + ds + sr^2 < 3d\,r \min(r, s). \tag{14}$$

Case $s \leq r$. Here $m_{\min} = rs$, and (14) reduces to $d(3rs - 2r - s) > sr^2$. When $3rs - 2r - s > 0$, the threshold is $d > sr^2/(3rs - 2r - s)$.

Case $s \geq r$. Here $m_{\min} = r^2$, and (14) reduces to $d(3r^2 - 2r - s) > sr^2$. For $s = O(r)$, this holds once $d = \Omega(r)$.

Balanced case $s = r = k$. The two counts become $N_{\text{Tucker}} = 3dk + k^3$ and $N_{\text{SwiGLU,eq}} = 3dk^2$, so Tucker is parameter-cheaper whenever

$$d > \frac{k^2}{3(k-1)}, \tag{15}$$

i.e. approximately $d > k/3$ for large k. In the transformer regime $d \gg k$,

$$\frac{N_{\text{SwiGLU,eq}}}{N_{\text{Tucker}}} = \frac{3dk^2}{3dk + k^3} = \frac{3dk}{3d + k^2} \approx k \quad (d \gg k^2), \tag{16}$$

giving a factor-k separation.

A.5. Numerical-rank intuition

Remark A.5 (Numerical-rank intuition). Theorem 4.2 is an exact-equality result. Under approximate simulation in (say) L^2 norm over a bounded input region, the rank lower bound is heuristically replaced by a numerical-rank condition: an aligned SwiGLU block needs at least as many units per gate j as the numerical rank of V_j above the target tolerance scale ε. We do not prove this; a complete approximation-theoretic version, including stability of the rank decomposition and the role of σ saturation in the gate, is left to future work.

B. Empirical details

B.1. Routing variance across depth

The Qwen2.5-0.5B routing coefficients $\alpha_j(x) = \sigma(g_j^\top x)$ are not approximately constant in any layer; per-channel $\operatorname{Var}_x[\alpha_j(x)]$ averaged across channels is U-shaped in depth, low in middle layers and rising sharply through the final layers (Figure 3). This complements the constant-α ablation (Table 1) and confirms that the gate carries non-trivial input-dependent signal at every depth, not just on average.

B.2. Stable rank of V_j

For the matched-budget Tucker LM trained from scratch (§B.5), the per-gate stable rank $\|V_j\|_F^2/\|V_j\|_{\text{op}}^2$ concentrates around $\bar{\rho} \approx 4$ across all gates and layers, with min 2.79 and max 5.30 (Figure 4). This is well above the rank-one ceiling that

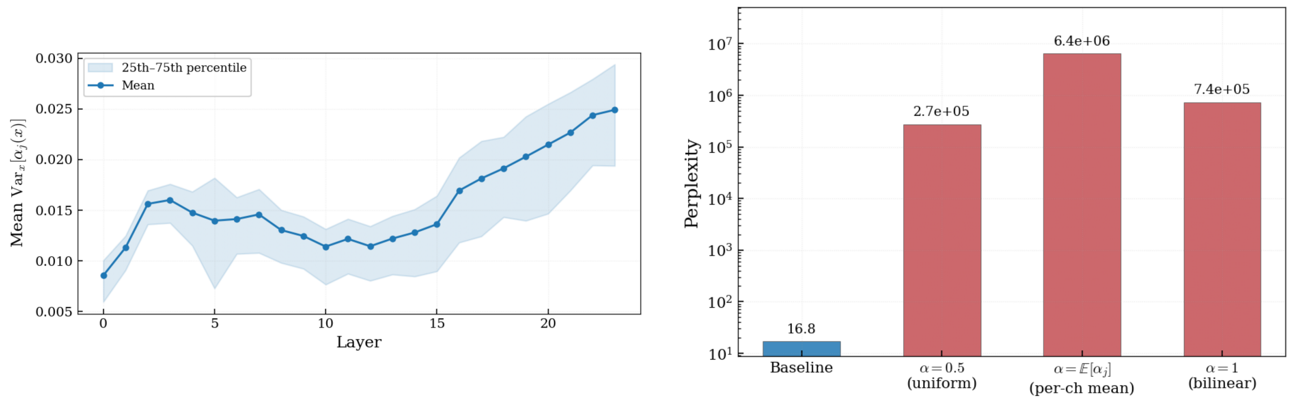

Figure 3. Routing validation on Qwen2.5-0.5B (Yang et al., 2024). Per-channel routing variance $\mathrm{Var}_x[\alpha_j(x)]$ averaged across channels is U-shaped in depth, low in middle layers and rising sharply through the final layers (24 SwiGLU layers, 4096-token chunk of WikiText-2). Right panel: validation perplexity under three constant-α ablations applied to all layers simultaneously, complementary to Table 1.

aligned SwiGLU at width r would impose (Theorem 4.2, ρ=1 point of Figure 2, right), and below the random-init Tucker baseline ($\bar{\rho} \approx 27.6$): the variance-preserving warm-start keeps the model close to a SwiGLU-equivalent diagonal subspace while still using cross-channel rank around $\bar{\rho} \approx 4$, well above one.

B.3. Distillation gap at matched budget

We isolate the layer-level expressivity gap from the end-to-end optimisation gap by distilling a single Tucker-LM layer (layer 4 of the trained Tucker model, d=512, r=128) into matched-parameter Tucker and SwiGLU students at four budget points (Figure 5). SwiGLU achieves lower MSE at small budgets (optimisation dominates), but at the matched-LM-config budget ($\sim$2.3M parameters, r=s=128 for Tucker, m=1493 for SwiGLU) Tucker achieves lower validation MSE than SwiGLU (8.38 vs. 1.08×10^1, a 22% reduction, mean over 3 seeds). The crossover at $\sim 7 \times 10^5$ parameters is consistent with the ρ-truncation curve of Figure 2.

B.4. Same-index pairing permutation

The same-index coupling among W, G, U in (4) is non-trivial: random permutation of any single factor matrix in a trained Qwen2.5-0.5B SwiGLU layer (8 seeds) degrades perplexity by 3–4 orders of magnitude in early layers and by 10–50% in middle layers, while the no-op control of permuting G, U, and W together by the same π recovers baseline to within 7.6×10^{-6} relative error (Figure 6). The asymmetry across depth is itself layer-dependent, but the no-op control rules out a unit-relabelling artefact: the model has learned a specific pairing of w_j, g_j, u_j at each hidden index j.

B.5. End-to-end LM training

Theorem 4.2 is a layer-level representational result; whether the gap survives end-to-end training is a separate question we cannot fully resolve at workshop scale. We train SwiGLU and Tucker-core LMs at matched parameter count ($\sim$50M params, 100M FineWeb-Edu tokens). Random Tucker initialisation pays a $\sim$30% perplexity penalty over SwiGLU, but a variance-preserving warm-start at the SwiGLU recovery point ($\mathcal{C}_{ooo} = 1$, off-diagonal std ε/r with $\varepsilon = 10^{-2}$, full-core std $1/r$) closes this gap (Table 2; loss curves in Figure 7). The -0.005 nats single-seed difference is sub-noise and we do not read it as Tucker outperforming SwiGLU; the takeaway is that the corrected initialisation matches SwiGLU end-to-end at this scale, leaving the architectural separation to be probed at the layer level (§5.3) and in distillation (Appendix B.3).

Validation loss vs. tokens for the SwiGLU and default-init Tucker runs of Table 2 is plotted in Figure 7. The variance-preserving Tucker run sits within the SwiGLU curve at every checkpoint and is omitted from the figure for legibility; final numbers are in Table 2.

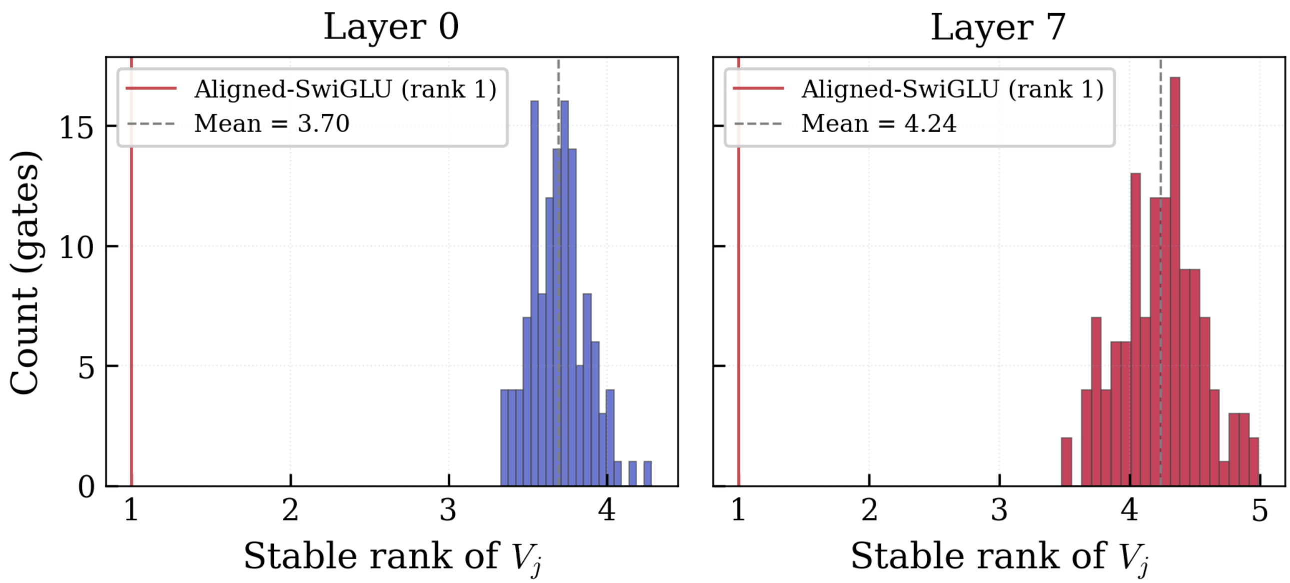

Figure 4. Per-gate stable rank $\|V_j\|_F^2/\|V_j\|_{\mathrm{op}}^2$ across gates and layers in a Tucker FFN trained from scratch with the variance-preserving warm start (d=512, r=s=128, L=8). Mean $\bar{\rho} \approx 3.97$ (min 2.79, max 5.30), well above the rank-one ceiling that aligned SwiGLU at width r would impose under Theorem 4.2 (red line).

Table 2. Validation loss after 100M FineWeb-Edu tokens at matched parameter count (d=512, L=8, r=s=128 for Tucker, m=1493 for SwiGLU; one seed). Variance-preserving init closes the optimisation gap between random-init Tucker and SwiGLU. Within-run std of val. loss over the last 10 checkpoints is ≈ 0.014 for both arches, comparable to or larger than the 0.005-nat gap, so the table should be read as a sanity check rather than a superiority claim.

arch / init	params	val_loss	ppl
SwiGLU	52.5M	4.758	116.5
Tucker, default init	52.5M	5.116	166.7
Tucker, var-preserving	52.5M	4.753	116.0

B.6. Cross-model robustness

The constant-α ablation result of §5.1 is not specific to Qwen2.5-0.5B. The same constant-α ablation pattern (4–7 orders of magnitude perplexity penalty) and the same depth profile of routing variance hold on Qwen3-1.7B (Yang et al., 2025), a separate model family at $\sim 3\times$ scale (Figure 8). We attempted Llama-3.2-1B first; the script falls back to Qwen3-1.7B when the Llama checkpoint is gated.

B.7. Hyperparameters and reproducibility

At the matched-parameter LM config (d=512, L=8), both SwiGLU and Tucker-core FFNs have $\approx 3.67\times10^7$ forward FLOPs per token, with measured throughput 112,496 vs. 75,009 tokens/sec on a single NVIDIA A100; Tucker is FLOPs-comparable, with the throughput gap reflecting the smaller-batch GEMM shapes the core contraction induces.

Tucker LM and SwiGLU LM training. We train both architectures from scratch on FineWeb-Edu `sample-10BT` (HuggingFace `HuggingFaceFW/fineweb-edu`, version pinned to the dataset hash on record at the time of the run) tokenized with the GPT-2 BPE tokenizer (vocabulary 50,257). Context length T=1024, batch size B=24 sequences (24K tokens per optimizer step). Optimizer: AdamW (β_1=0.9, β_2=0.95), weight decay 0.1 on all non-bias, non-RMSNorm parameters; the Tucker core C is held in a separate parameter group with weight decay 0. Gradient clipping at 1.0. LR schedule: linear warmup over 1% of total steps, cosine decay to 1% of peak; peak LR 0.0003. Precision: bf16 autocast for the forward/backward, fp32 master parameters and optimizer state. Hardware: single NVIDIA A100 (80 GB). Total: ≈ 6000 optimizer steps, 147M training tokens. We log validation cross-entropy every $\approx$ 2M tokens on a held-out $\approx$ 2M-token slice of FineWeb-Edu (disjoint from the training stream by random seed); the curves in Figure 7 use these checkpoints.

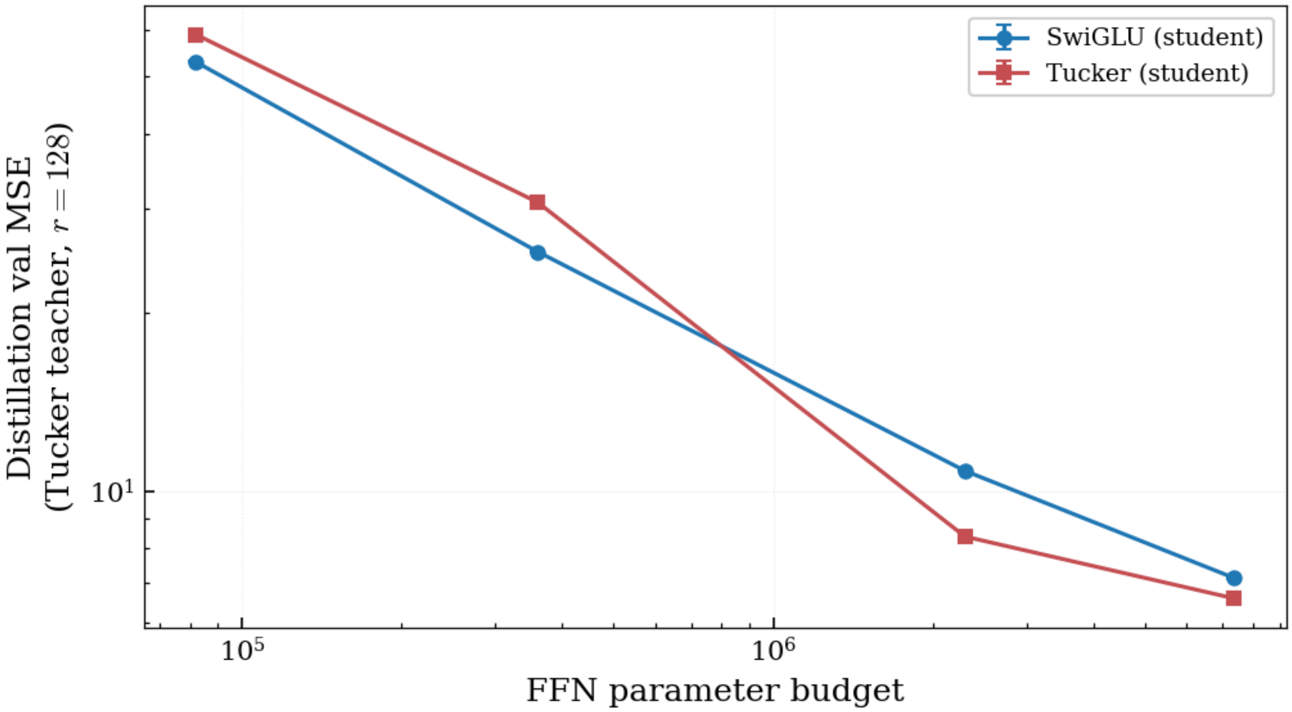

Figure 5. Distillation validation MSE (Tucker teacher, r=128) for SwiGLU and Tucker FFN students fit to a trained Tucker teacher layer (layer 4 of the trained Tucker LM, d=512; matched parameter budgets, three seeds per budget). At small budgets, optimization dominates and SwiGLU wins; at the matched-LM-config budget ($\sim$ 2.3M parameters: r=s=128 for Tucker, m=1493 for SwiGLU) Tucker matches the teacher 22% more accurately (1.08×10^1 vs. 8.38 val MSE). Crossover near 7×10^5 parameters.

The variance-preserving Tucker initialization is $C_{oij} \sim \mathcal{N}(0, 1/r^2)$ for the off-diagonal entries with diagonal warm start $C_{ooo} \leftarrow 1$ (using o for the core output index to avoid collision with the routing scalar $\alpha_j(x)$), so that at initialization the layer evaluates the SwiGLU recovery form $z_o = p_o\,\mathrm{SiLU}(q_o)$ plus $O(\varepsilon/r)$ noise at ε=10^{-2}. Latent projections P, Q and output projection R use He-uniform init.

Constant-α ablation. Calibration set: 32 sequences of length 1024 from the WikiText-2 train split (32K tokens), used to estimate the per-channel mean $\bar{\alpha}_j = \mathbb{E}_x[\alpha_j(x)]$ for the $\alpha = \bar{\alpha}$ condition. Activations are tapped at the input to the gate projection (*after* the FFN's input RMSNorm, which is the standard tap point in HuggingFace SwiGLU implementations). All 24 layers are ablated simultaneously by replacing $\sigma(g_j^\top x)$ with the constant for that condition. Evaluation perplexity: 4,096-token chunk of WikiText-2 validation.

Synthetic Tucker teacher (Figure 1). d=64. Per teacher, $P, Q, R \sim \mathcal{N}(0, 1/d)$; $C \in \mathbb{R}^{k\times k\times k}$ is sampled $C_{oij} \sim \mathcal{N}(0, 1)$ and resampled until every frontal slice $C^{(j)} = C[:, :, j]$ has full rank k (typically the first sample suffices). Inputs $x \sim \mathcal{N}(0, I_d)$; train/val sizes 50000/5000. Adam, lr 0.003, batch 512, 8000 steps with cosine schedule (1% floor, no warmup). 8 seeds per (k, m) cell, val MSE reported as the minimum across seeds (best fit achievable). Aligned-SwiGLU students use the teacher's P, Q as fixed latent dictionaries (the matched-coordinates assumption of Theorem 4.2); each unit's gate index is assigned round-robin over the k gates. The SVD-construction marker (Figure 1) is computed analytically from the teacher's $V_j = RC^{(j)}$ via SVD without any optimization.

Distillation (Figure 5). Teacher: layer ℓ=4 of a trained Tucker LM (d=512, r=s=128). Inputs are sampled by streaming FineWeb-Edu through the teacher LM and capturing the residual-stream activations entering the chosen FFN; train/val sizes 80000/8000 tokens. Adam, lr 0.001, batch 512, 8000 steps with cosine schedule (1% floor, 5% warmup) and gradient clipping 1.0. 3 seeds per budget; error bars are ±one std across seeds.

Code. Code will be released upon acceptance.

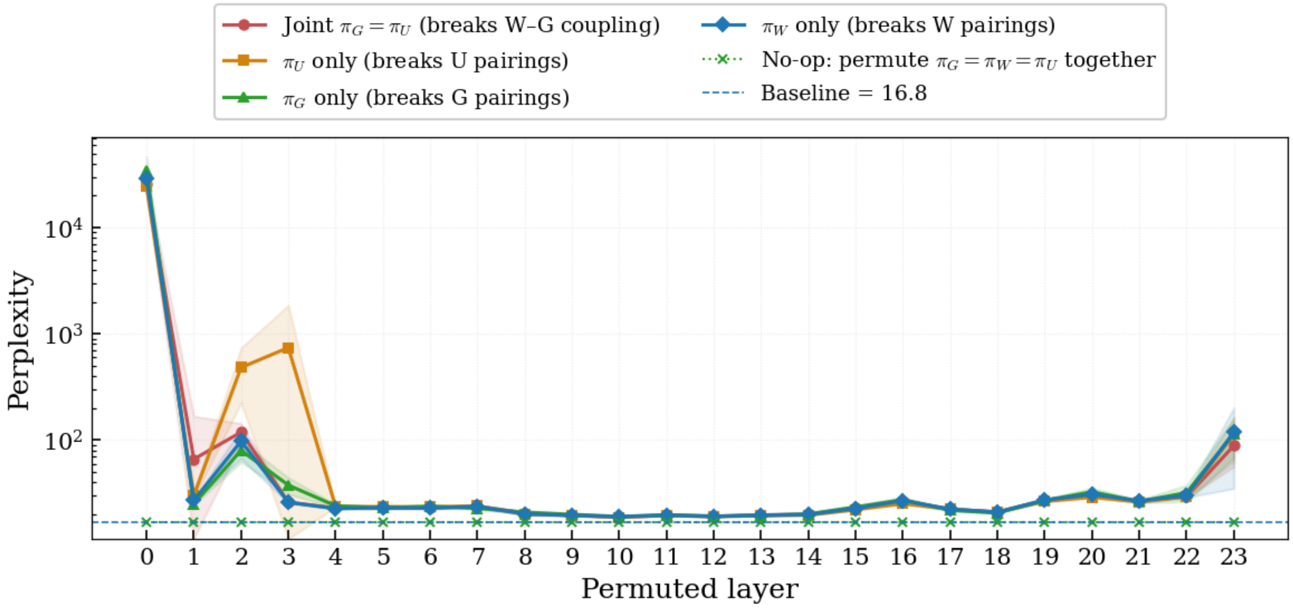

Figure 6. Per-layer perplexity under random permutations of one trained SwiGLU layer in Qwen2.5-0.5B (eight seeds). Joint $\pi_G{=}\pi_U$ breaks the same-index W–G coupling; π_U-only is a control that breaks U pairing alone. The no-op control (joint $\pi_G{=}\pi_U{=}\pi_W$) overlays the baseline at every layer to within 7.6×10^{-6} relative error, ruling out a unit-relabelling artifact.

C. Interpretation of the diagonal bottleneck

The theorem isolates the cost of diagonal pairing. A Tucker-GLU block parameterizes cross-channel interactions $p_i \leftrightarrow \phi(q_j)$ through the dense core entries $\mathcal{C}_{oij}$; after applying the output factor R, each gate q_j acquires an output-by-main-feature matrix $V_j \in \mathbb{R}^{d\times r}$. An aligned SwiGLU block can realize the same map only by decomposing each V_j as a sum of rank-one matrices, with one hidden unit per rank-one term. This forces $m_{\min} = \sum_j \operatorname{rank}(V_j)$, generically $r\min(r,s)$, against the Tucker parameter count $O(d(r+s)+sr^2)$. In the balanced regime $r=s=k$, this is a factor-k blowup. The diagonal restriction in SwiGLU is thus not harmless at matched latent dictionaries; it converts cross-channel information stored in $O(k^3)$ core entries into $O(dk^2)$ aligned-SwiGLU rank-one atoms.

Connections to circuits and transcoders. The routed-CP / Tucker view sits alongside circuit / tensor-factorization formalisms (Loconte et al., 2024) and matches the input-invariant-atom + input-dependent-routing factorization that transcoders (Dunefsky et al., 2024; Paulo et al., 2025) arrive at empirically; we leave a formal connection to future work.

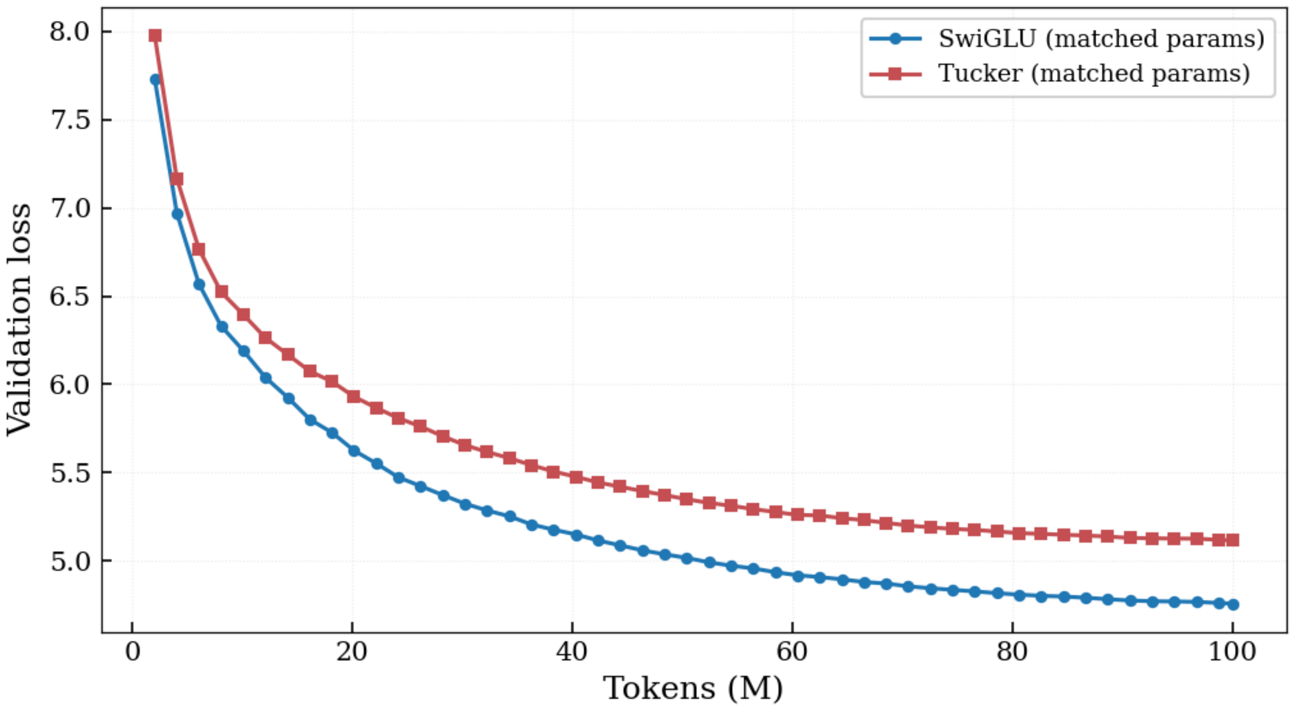

Figure 7. Validation loss curves for the SwiGLU and default-init Tucker runs of Table 2, trained from scratch on FineWeb-Edu (Penedo et al., 2024) at matched parameter count (d=512, L=8, ∼50M params, 100M tokens). The variance-preserving Tucker run sits within the SwiGLU curve at every checkpoint and is omitted for legibility; final numbers are in Table 2.

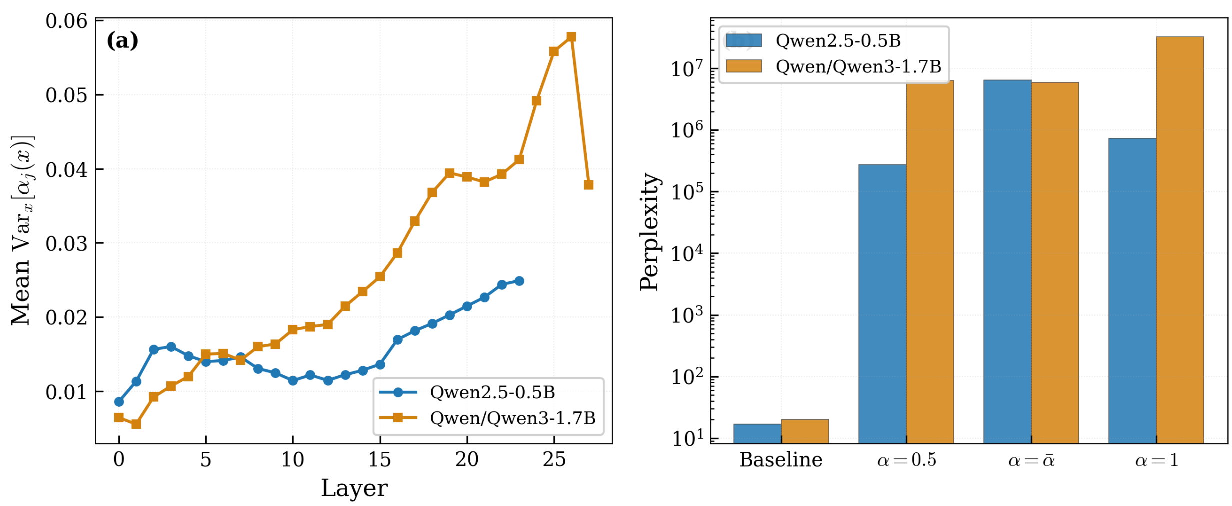

Figure 8. Cross-model robustness of the routed-CP picture on Qwen3-1.7B (Yang et al., 2025) overlaid on the Qwen2.5-0.5B baseline (Yang et al., 2024). (a) Per-layer mean routing variance $\mathrm{Var}_x[\alpha_j(x)]$. (b) Constant-$\alpha$ ablation perplexity (baseline, uniform, per-channel mean, α=1); the larger model relies even more heavily on routing, with an α=1 over baseline ratio of 1.62×10^6 vs. 4.38×10^4 for Qwen2.5-0.5B. Both model families exhibit the same qualitative pattern, ruling out a Qwen2.5-specific artifact.